Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

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  - series and parallel connections, source transformation

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

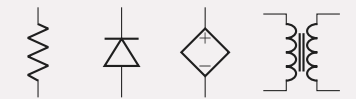
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

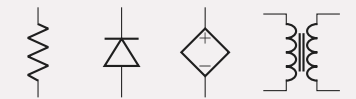


later: the ideal op-amp

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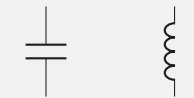
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Its law has a derivative.

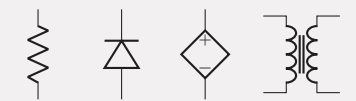


also coupled inductors

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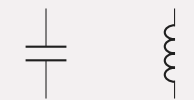
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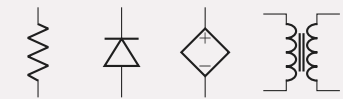
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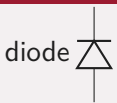
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

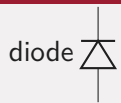


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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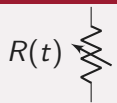
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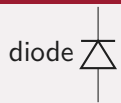


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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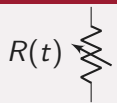
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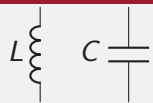
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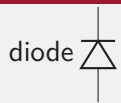


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

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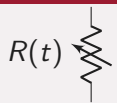
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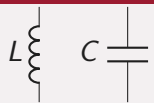
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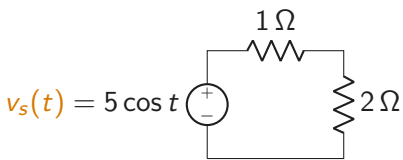
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LTI resistive circuit = linear + time-invariant + resistive:

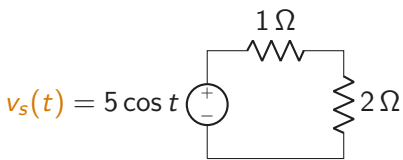
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

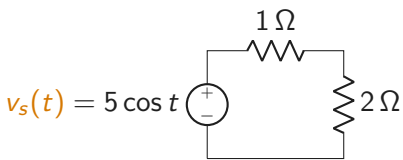
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The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

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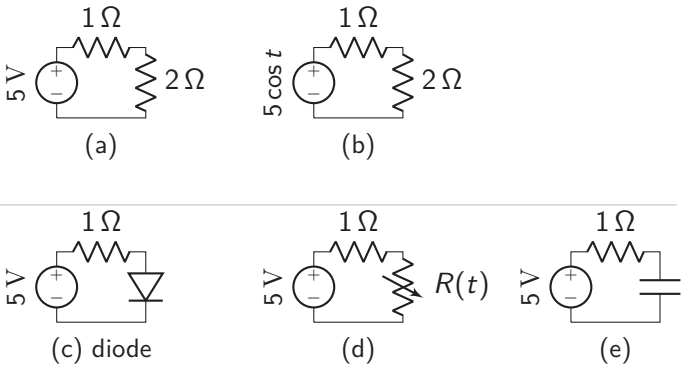
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

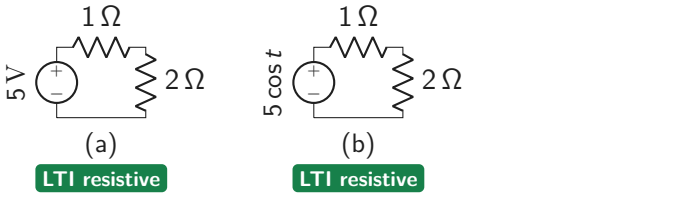
Which circuit is which?



A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

What we gain

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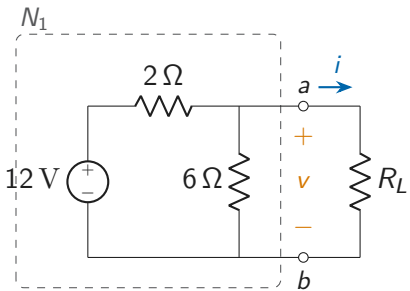
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- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

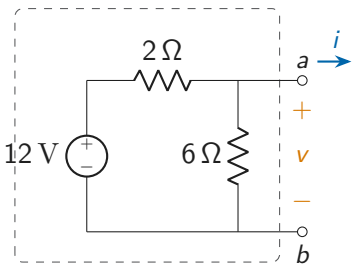
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A first example: what does R_L see?



R_L (the load) can be any resistive element.
Find v and i.

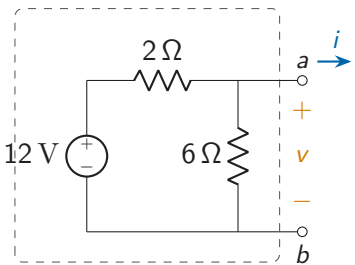
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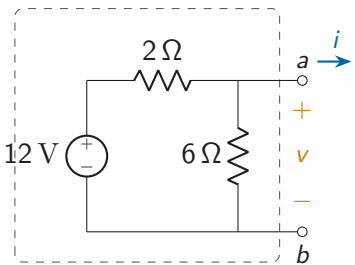


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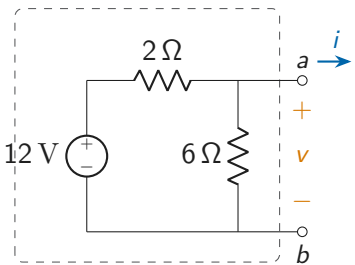
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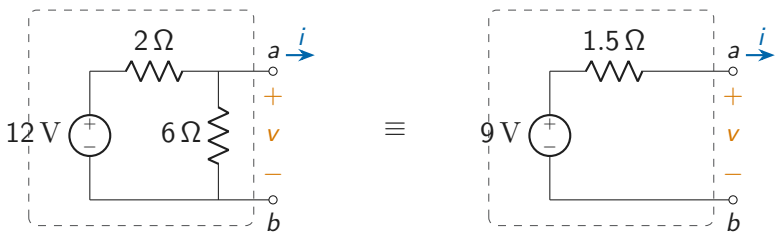
Solve for v: $v = 9 - 1.5\,i$

Read the result as a circuit

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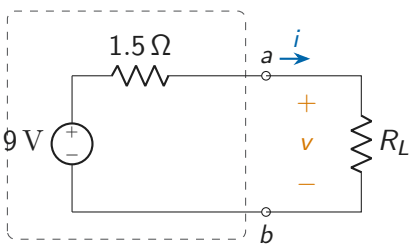
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The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

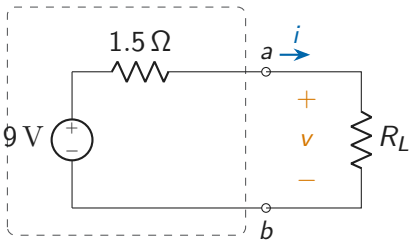
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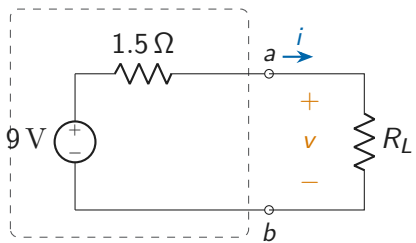
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Connect any load R_L to the simple circuit.

The $1.5\,\Omega$ and R_L are in series: a voltage divider.

$$v = \frac{R_L}{1.5 + R_L} \cdot 9\,\mathrm{V}, \qquad i = \frac{v}{R_L} = \frac{9}{1.5 + R_L}$$

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For $R_L = 3\,\Omega$: $\quad v = \dfrac{3}{4.5} \cdot 9 = 6\,\mathrm{V}, \quad i = 2\,\mathrm{A}.$

What did we just do?

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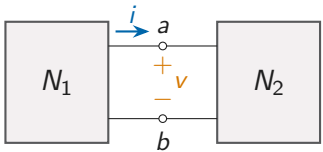
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question	**answered by**
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

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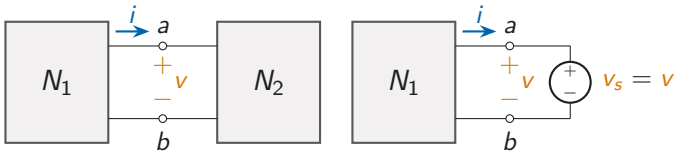
The substitution theorem



(I)

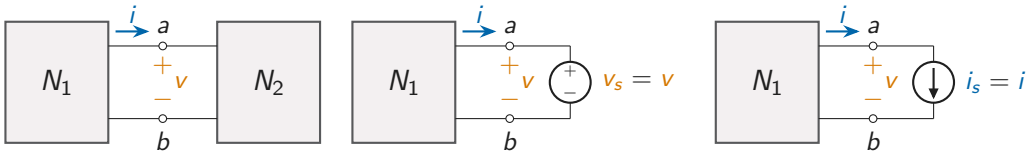
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



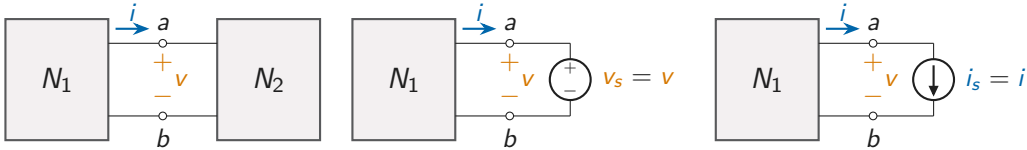
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



... or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Condition: each circuit has a unique solution.

A unique solution

A circuit with n nodes and b branches has $2b$ unknowns, $v_1, \ldots, v_b$ and $i_1, \ldots, i_b$, and $2b$ equations:

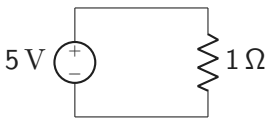
KCL	$A\mathbf{i} = 0$	$n-1$
KVL	$B\mathbf{v} = 0$	$b-n+1$
element laws	one for each branch	b

Definition

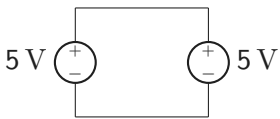
The circuit has a unique solution if exactly one $(\mathbf{v}, \mathbf{i})$ satisfies all $2b$ equations.

How do we check it? For an LTI resistive circuit: see the next slides.

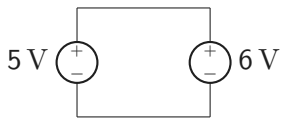
A unique solution? Three small circuits



(a)

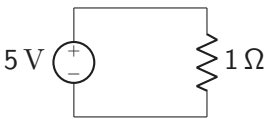


(b)



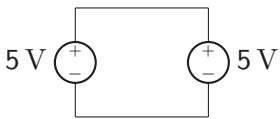
(c)

A unique solution? Three small circuits



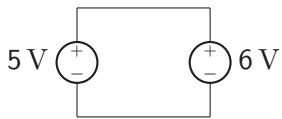
(a)

✓ unique: $i = 5\,\mathrm{A}$



(b)

× many: any current



(c)

× none: KVL fails

A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

T, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. u: the source values.

The solution is unique exactly when T is invertible: $\det T \neq 0$.

Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

T, the coefficient matrix: one row per equation, holding the numbers that multiply the unknowns. u: the source values.

The solution is unique exactly when T is invertible: $\det T \neq 0$.

(a) $5\,\text{V}$  $1\,\Omega$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix}$$

KCL
KVL
source 1
resistor 2

$\det T = -1$: ✓ unique

Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

A unique solution: in matrix form

Collect the $2b$ unknowns in one vector $x = (v_1, \ldots, v_b, i_1, \ldots, i_b)$. In an LTI resistive circuit all $2b$ equations are linear: $T\,x = u$.

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The solution is unique exactly when T is invertible: $\det T \neq 0$.

(a) $5\,\mathrm{V}$, $1\,\Omega$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 0 \end{bmatrix} \begin{matrix} \text{KCL} \\ \text{KVL} \\ \text{source 1} \\ \text{resistor 2} \end{matrix}$$

$\det T = -1$: ✓ unique

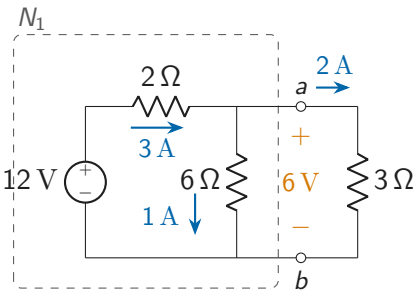
(b) $5\,\mathrm{V}$, $5\,\mathrm{V}$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \\ 5 \end{bmatrix} \begin{matrix} \text{KCL} \\ \text{KVL} \\ \text{source 1} \\ \text{source 2} \end{matrix}$$

$\det T = 0$: × i_1, i_2 appear only in KCL

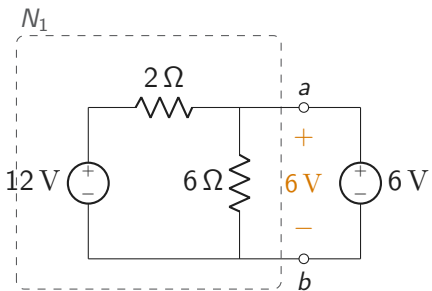
Branch 1: the left source; branch 2: the right element; both oriented from top to bottom.

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

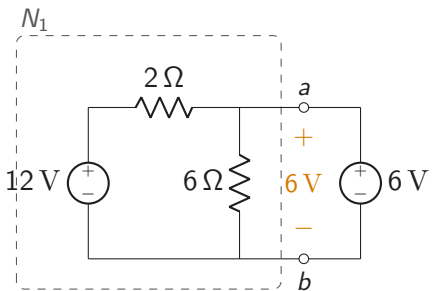
(I) and (II): an example



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(II) Replace R_L by a $6\,\mathrm{V}$ source. What are the values now?

(I) and (II): an example



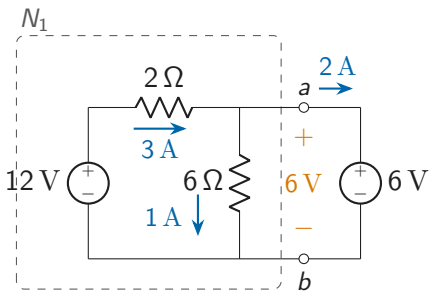
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KCL at a with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

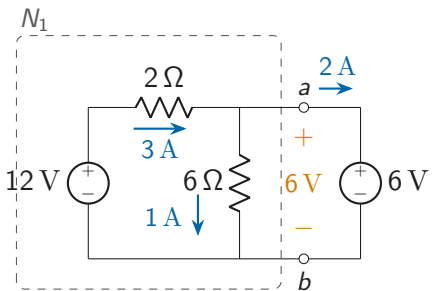
(II) Replace R_L by a $6\,\mathrm{V}$ source. What are the values now?

KCL at a with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

(II) Replace R_L by a $6\,\mathrm{V}$ source. What are the values now?

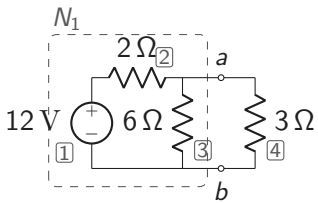
KCL at a with $v = 6\,\mathrm{V}$:

$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

In (I) we solved the same equation: there too $v = 6\,\mathrm{V}$.

(I) and (II): the example in matrix form

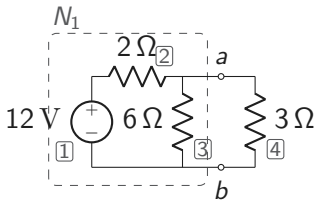


Branch 1 from + to −, branch 2 to the right, branches 3, 4 downwards.

(I): branch 4 is the 3 Ω.

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -3 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_1 \\ i_2 \\ i_3 \\ i_4 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 12 \\ 0 \\ 0 \\ 0 \end{array}\right] \begin{array}{l} \text{KCL, top left} \\ \text{KCL, node } a \\ \text{KVL, left loop} \\ \text{KVL, right loop} \\ 12\,\text{V source} \\ 2\,\Omega \\ 6\,\Omega \\ \text{branch 4} \end{array}$$

(I) and (II): the example in matrix form

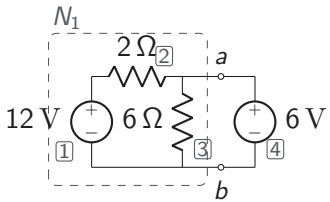


Branch 1 from + to −, branch 2 to the right, branches 3, 4 downwards.

Question. In (II), which row changes?

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & -3 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_1 \\ i_2 \\ i_3 \\ i_4 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 12 \\ 0 \\ 0 \\ 0 \end{array}\right] \begin{array}{l} \text{KCL, top left} \\ \text{KCL, node } a \\ \text{KVL, left loop} \\ \text{KVL, right loop} \\ 12\,\text{V source} \\ 2\,\Omega \\ 6\,\Omega \\ \text{branch 4} \end{array}$$

(I) and (II): the example in matrix form



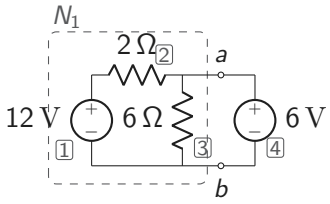
Branch 1 from + to −, branch 2 to the right, branches 3, 4 downwards.

(II): only the last row changes, $v_4 = 6$.

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_1 \\ i_2 \\ i_3 \\ i_4 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 12 \\ 0 \\ 0 \\ 6 \end{array}\right]$$

KCL, top left
KCL, node a
KVL, left loop
KVL, right loop
12 V source
2 Ω
6 Ω
branch 4

(I) and (II): the example in matrix form



Branch 1 from + to −, branch 2 to the right, branches 3, 4 downwards.

Put in the solution of (I):

$x^* = (12, 6, 6, 6, -3, 3, 1, 2)$

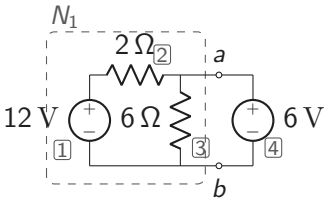
rows 1–7: as in (I) ✓

row 8: $v_4^* = 6$ ✓

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_1 \\ i_2 \\ i_3 \\ i_4 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 12 \\ 0 \\ 0 \\ 6 \end{array}\right]$$

KCL, top left
KCL, node a
KVL, left loop
KVL, right loop
12 V source
2 Ω
6 Ω
branch 4

(I) and (II): the example in matrix form



Branch 1 from + to −, branch 2 to the right, branches 3, 4 downwards.

Put in the solution of (I):

$x^* = (12, 6, 6, 6, -3, 3, 1, 2)$

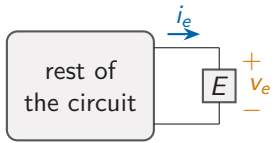
rows 1–7: as in (I) ✓

row 8: $v_4^* = 6$ ✓

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 \\ -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ i_1 \\ i_2 \\ i_3 \\ i_4 \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \\ 12 \\ 0 \\ 0 \\ 6 \end{array}\right] \begin{array}{l} \text{KCL, top left} \\ \text{KCL, node } a \\ \text{KVL, left loop} \\ \text{KVL, right loop} \\ 12\,\text{V source} \\ 2\,\Omega \\ 6\,\Omega \\ \text{branch 4} \end{array}$$

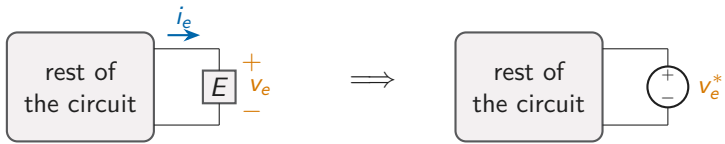
$\det T_{\text{II}} = -12 \neq 0$: x^* is the only solution of (II).

(I) and (II): one element first



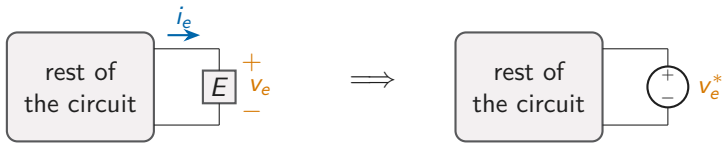
Any element E of a circuit; in the solution of (I) it has v_e^* and i_e^*.

(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

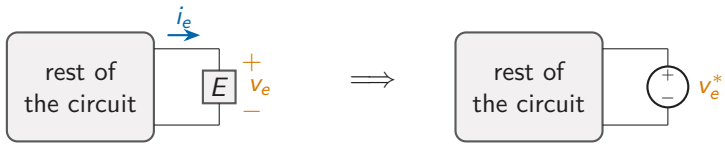
(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)
KCL	$A\,\mathbf{i} = 0$
KVL	$B\,\mathbf{v} = 0$
other $b-1$ elements	their laws
element E	the law of E

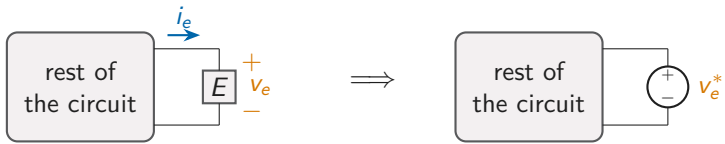
(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	**(II)**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	(II)
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of E included.

(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	**(II)**	$(\mathbf{v}^*, \mathbf{i}^*)$ **in (II)?**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (II)?
KCL	$A\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.

(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (II)?
KCL	$A\,\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.

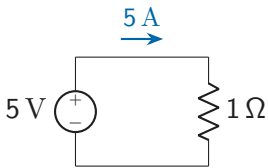
(I) and (II): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (II)?

	(I)	(II)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (II)?
KCL	$A\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$v_e = v_e^*$	✓ $v_e^* = v_e^*$

- A voltage source takes any current, so $i_e = i_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (II); it has only one solution, so nothing changes, also $i_e = i_e^*$.
- This needs (II) to have one solution; (I) having one is not enough (next slide).

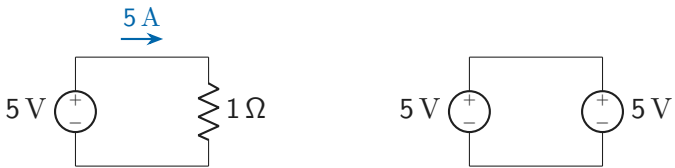
(I) and (II): why (II) must also have one solution



(I)

(I): a $5\,\mathrm{V}$ source and a $1\,\Omega$. One solution: $v = 5\,\mathrm{V}$, $i = 5\,\mathrm{A}$.

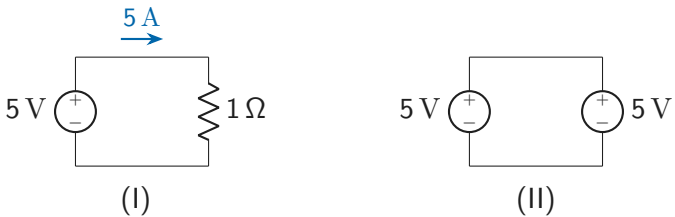
(I) and (II): why (II) must also have one solution



(I): a $5\,\mathrm{V}$ source and a $1\,\Omega$. One solution: $v = 5\,\mathrm{V}$, $i = 5\,\mathrm{A}$.

(II): replace the $1\,\Omega$ by a source equal to its voltage, $5\,\mathrm{V}$.

(I) and (II): why (II) must also have one solution

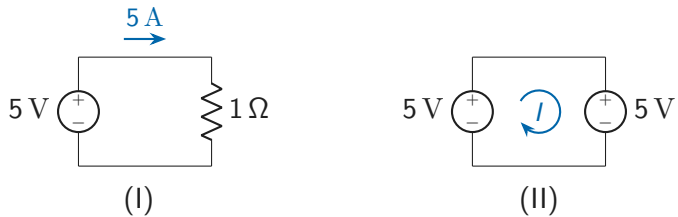


(I): a $5\,\mathrm{V}$ source and a $1\,\Omega$. One solution: $v = 5\,\mathrm{V}$, $i = 5\,\mathrm{A}$.

(II): replace the $1\,\Omega$ by a source equal to its voltage, $5\,\mathrm{V}$.

Question. Does (II) have one solution?

(I) and (II): why (II) must also have one solution



(I): a $5\,\mathrm{V}$ source and a $1\,\Omega$. One solution: $v = 5\,\mathrm{V}$, $i = 5\,\mathrm{A}$.

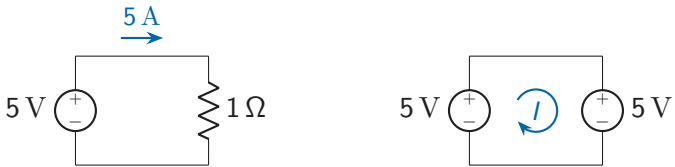
(II): replace the $1\,\Omega$ by a source equal to its voltage, $5\,\mathrm{V}$.

Question. Does (II) have one solution?

No: KVL holds for any current I around the loop.

Both sources take any current.

(I) and (II): why (II) must also have one solution



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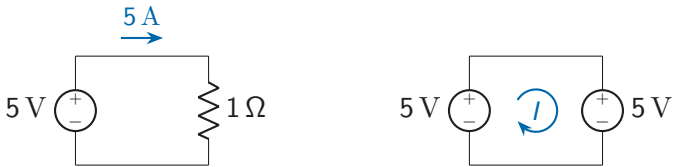
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$I = 5\,\mathrm{A}$ is one solution of (II), but not the only one.

(I) and (II): why (II) must also have one solution



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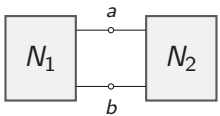
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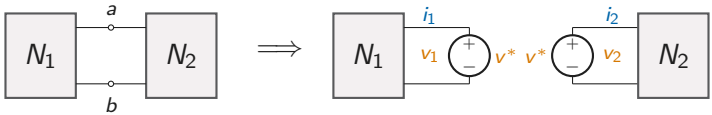
So the theorem asks that (II) also has one solution.

(I) and (II): divide N_1 and N_2



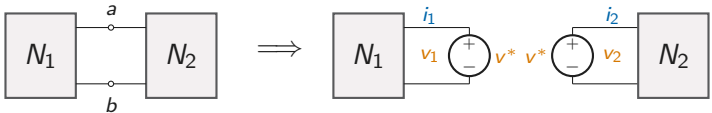
(I): N_1 and N_2 joined at a, b; the solution has v^*, i^* at the port.

(I) and (II): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

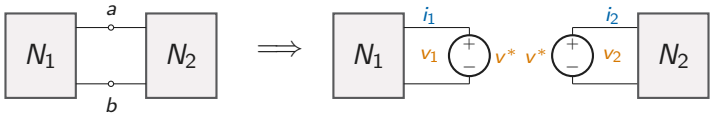
(I) and (II): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

	(I)	**divided**
inside N_1	KCL, KVL, element laws	the same
inside N_2	KCL, KVL, element laws	the same
the connection	$v_1 = v_2, \;\; i_1 = i_2$	$v_1 = v^*, \;\; v_2 = v^*$

(I) and (II): divide N_1 and N_2

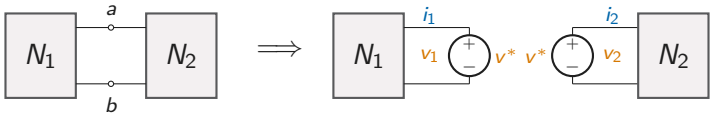


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- The solution of (I) satisfies all these equations, since $v_1 = v_2 = v^*$. ✓

(I) and (II): divide N_1 and N_2

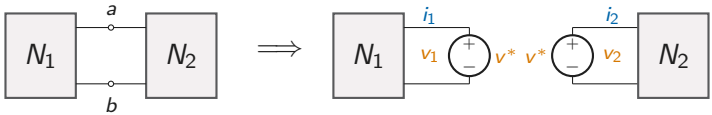


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- One solution, so this is it: nothing changes in either part.

(I) and (II): divide N_1 and N_2

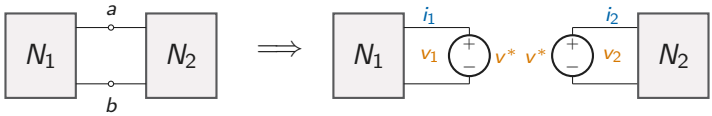


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(I) and (II): divide N_1 and N_2

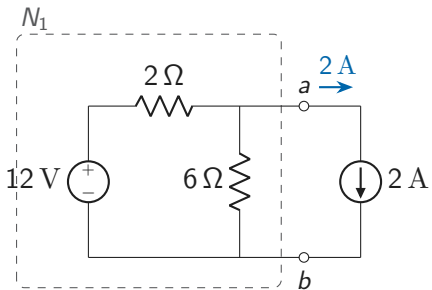


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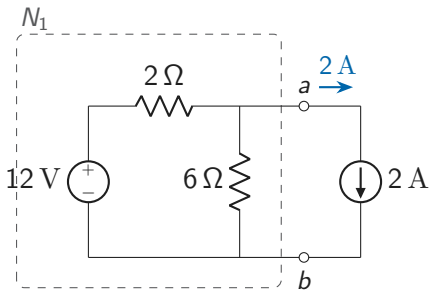
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- One solution, so this is it: nothing changes in either part.
- The left part is exactly (II): nothing inside N_1 changes, also $i_1 = i^*$.
- N_2 is not needed at all: it can be anything.

(I) and (III): an example



(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

(I) and (III): an example

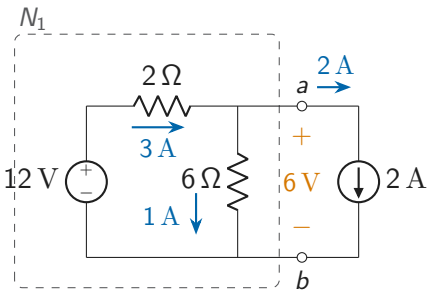


(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

KCL at a with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

(I) and (III): an example



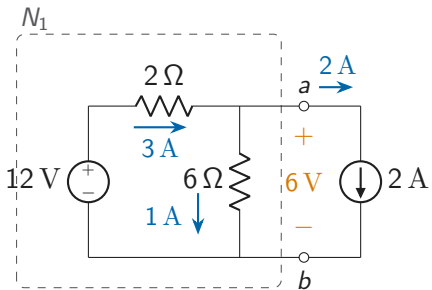
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$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

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(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

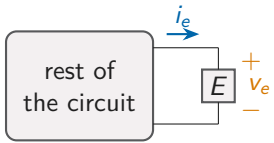
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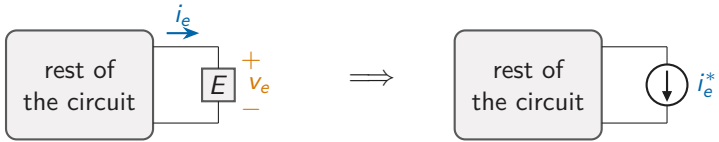
In (I) this equation holds too: there $i = 2\,\mathrm{A}$.

(I) and (III): one element first



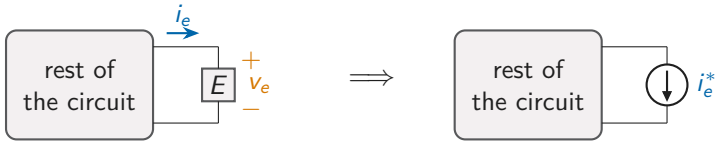
Any element E of a circuit; in the solution of (I) it has v_e^* and i_e^*.

(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

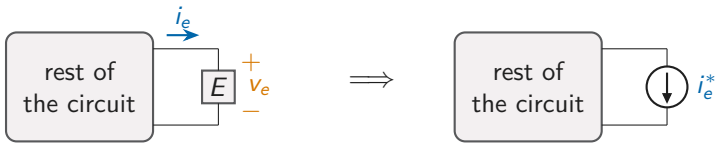
(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

	(I)
KCL	$A\,\mathbf{i} = 0$
KVL	$B\,\mathbf{v} = 0$
other $b-1$ elements	their laws
element E	the law of E

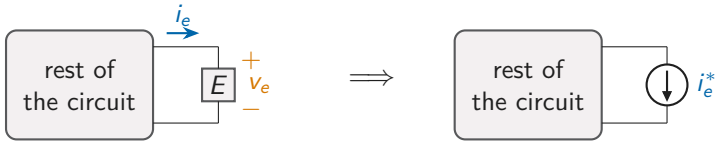
(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

	(I)	**(III)**
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$i_e = i_e^*$

(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

	(I)	(III)
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$i_e = i_e^*$

- The solution of (I), $(\mathbf{v}^*, \mathbf{i}^*)$, satisfies every equation of (I), the law of E included.

(I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

	(I)	(III)
KCL	$A\,\mathbf{i} = 0$	the same (same graph)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)
other $b-1$ elements	their laws	the same
element E	the law of E	$i_e = i_e^*$

(I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

	(I)	(III)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (III)?
KCL	$A\,\mathbf{i} = 0$	the same (same graph)	✓ as in (I)
KVL	$B\,\mathbf{v} = 0$	the same (same graph)	✓ as in (I)
other $b-1$ elements	their laws	the same	✓ as in (I)
element E	the law of E	$i_e = i_e^*$	✓ $i_e^* = i_e^*$

(I) and (III): one element, check the old solution

Does $(\mathbf{v}^*, \mathbf{i}^*)$ also satisfy the equations of (III)?

	(I)	(III)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (III)?
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element E	the law of E	$i_e = i_e^*$	✓ $i_e^* = i_e^*$

- A current source takes any voltage, so $v_e = v_e^*$ is allowed.

(I) and (III): one element, check the old solution

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	(I)	(III)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (III)?
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- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.

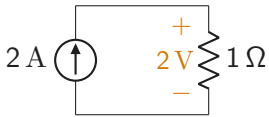
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	(I)	(III)	$(\mathbf{v}^*, \mathbf{i}^*)$ in (III)?
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- A current source takes any voltage, so $v_e = v_e^*$ is allowed.
- So $(\mathbf{v}^*, \mathbf{i}^*)$ solves (III); it has only one solution, so nothing changes, also $v_e = v_e^*$.
- This needs (III) to have one solution; (I) having one is not enough (next slide).

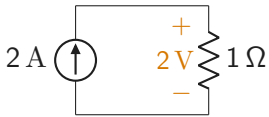
(I) and (III): why (III) must also have one solution



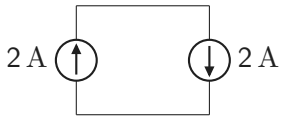
(I)

(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

(I) and (III): why (III) must also have one solution



(I)

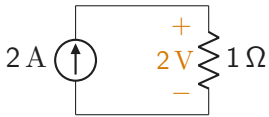


(III)

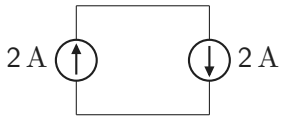
(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

(III): replace the $1\,\Omega$ by a source equal to its current, $2\,\mathrm{A}$.

(I) and (III): why (III) must also have one solution



(I)



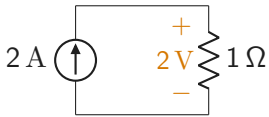
(III)

(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

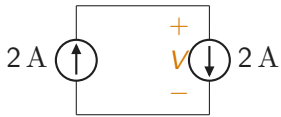
(III): replace the $1\,\Omega$ by a source equal to its current, $2\,\mathrm{A}$.

Question. Does (III) have one solution?

(I) and (III): why (III) must also have one solution



(I)



(III)

(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

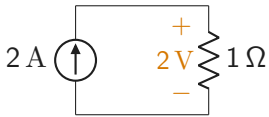
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Question. Does (III) have one solution?

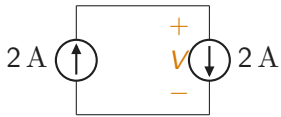
No: KCL holds for any voltage V across the two sources.

Both sources take any voltage.

(I) and (III): why (III) must also have one solution



(I)



(III)

(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

(III): replace the $1\,\Omega$ by a source equal to its current, $2\,\mathrm{A}$.

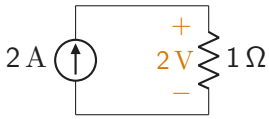
Question. Does (III) have one solution?

No: KCL holds for any voltage V across the two sources.

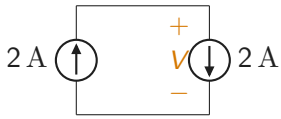
Both sources take any voltage.

$V = 2\,\mathrm{V}$ is one solution of (III), but not the only one.

(I) and (III): why (III) must also have one solution



(I)



(III)

(I): a $2\,\mathrm{A}$ source and a $1\,\Omega$. One solution: $i = 2\,\mathrm{A}$, $v = 2\,\mathrm{V}$.

(III): replace the $1\,\Omega$ by a source equal to its current, $2\,\mathrm{A}$.

Question. Does (III) have one solution?

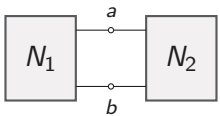
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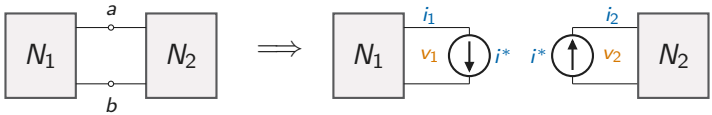
So the theorem asks that (III) also has one solution.

(I) and (III): divide N_1 and N_2



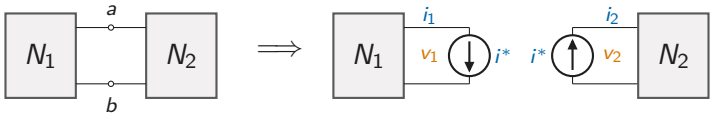
(I): N_1 and N_2 joined at a, b; the solution has v^*, i^* at the port.

(I) and (III): divide N_1 and N_2



Divide them and give each part its own source. Assume a unique solution.

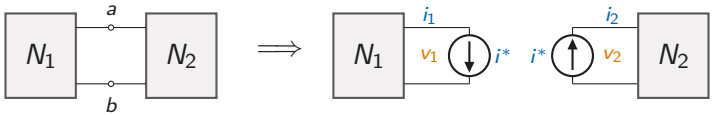
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Divide them and give each part its own source. Assume a unique solution.

	(I)	**divided**
inside N_1	KCL, KVL, element laws	the same
inside N_2	KCL, KVL, element laws	the same
the connection	$v_1 = v_2, \;\; i_1 = i_2$	$i_1 = i^*, \;\; i_2 = i^*$

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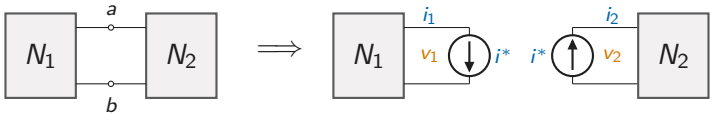


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- The solution of (I) satisfies all these equations, since $i_1 = i_2 = i^*$. ✓

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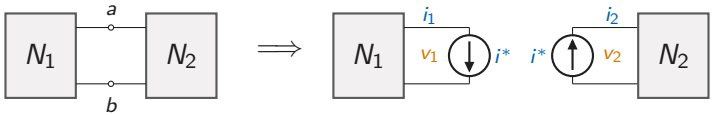


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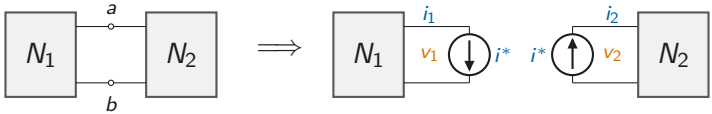


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- The left part is exactly (III): nothing inside N_1 changes, also $v_1 = v^*$.
- N_2 is not needed at all: it can be anything.

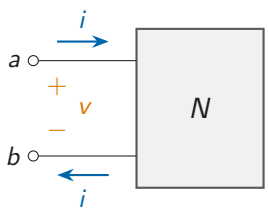
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Ports and one-ports

Port, one-port

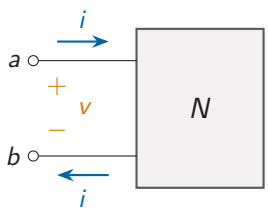
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



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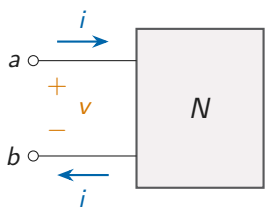


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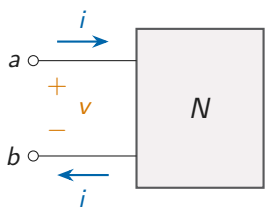


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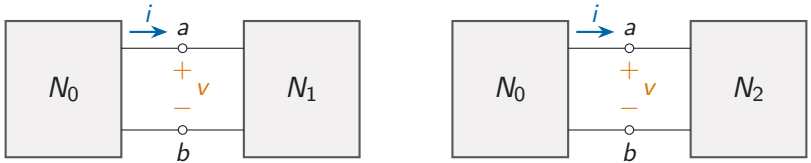


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

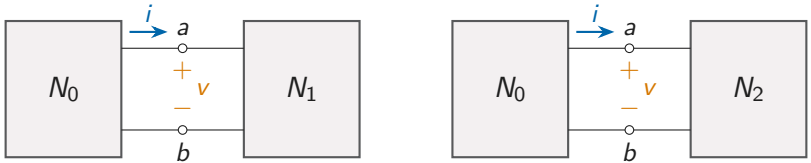
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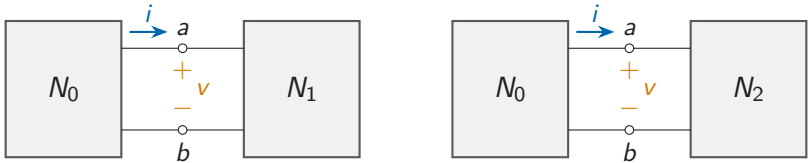


For any N_0: every branch voltage and current inside N_0 is the same in both.

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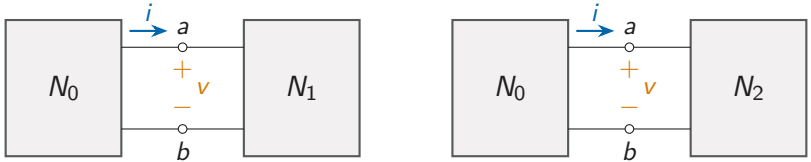
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Why: the equations are those of N_0 plus the i–v characteristic at the port.

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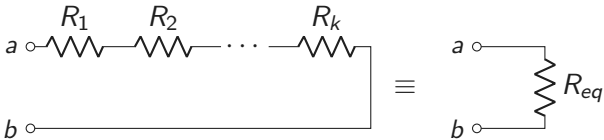
Inside N_1 and N_2 the branch values may differ.

Roadmap

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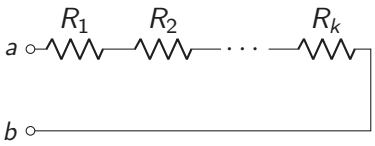
Series connection

Same current i through every element (KCL); the voltages add (KVL).

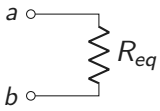


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$\equiv$

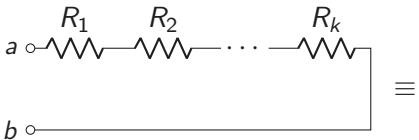


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

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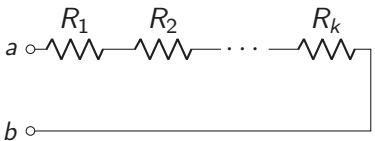


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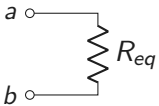
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

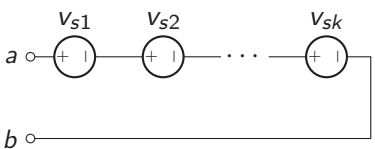
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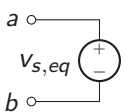
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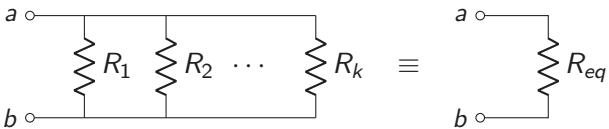


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

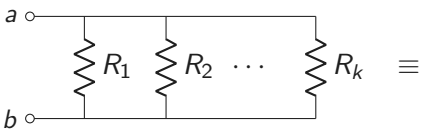
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



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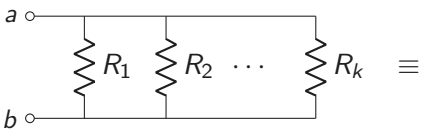


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

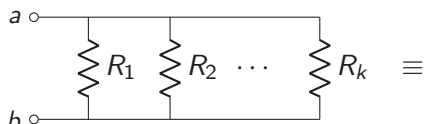


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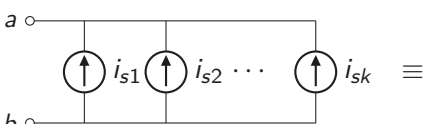
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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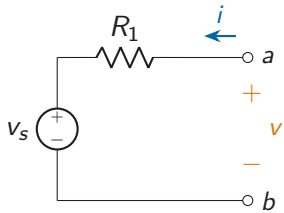
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

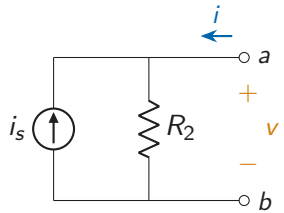
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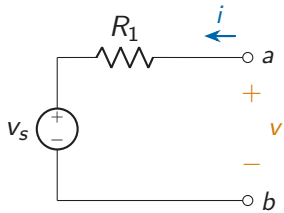
One source and one resistor

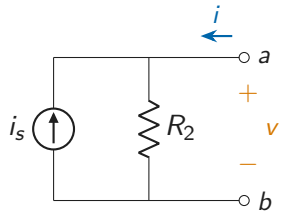




When do these two one-ports have the same i–v characteristic?

One source and one resistor

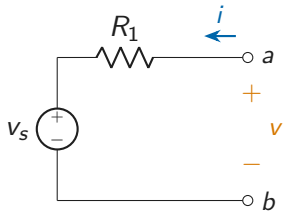




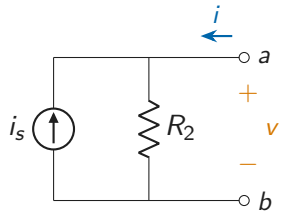
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



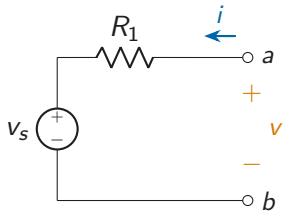
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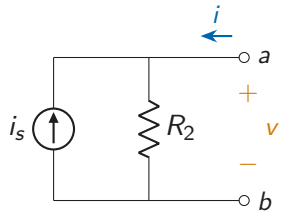
$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

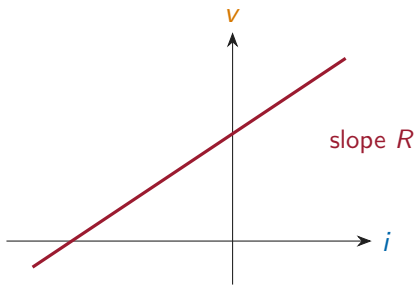
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



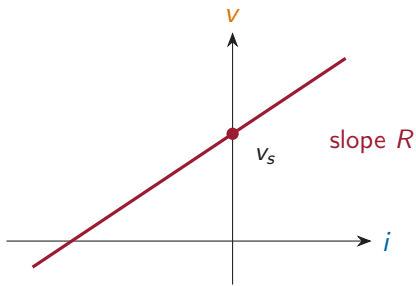
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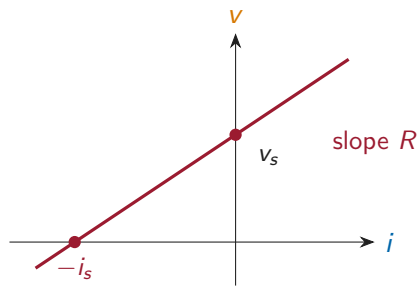
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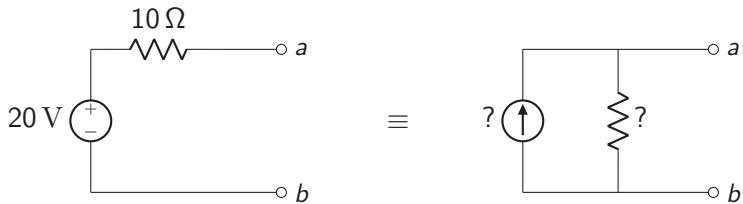
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

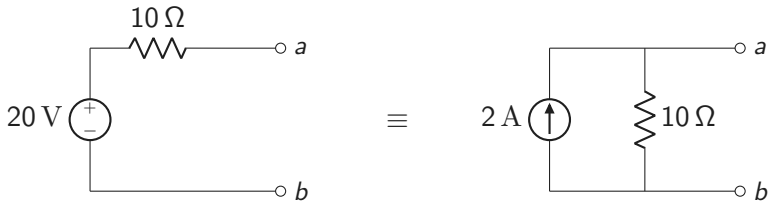


Example: a source transformation



Find the current source and the resistor.

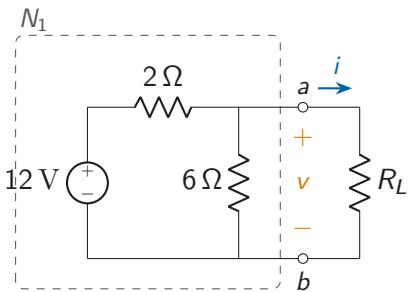
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

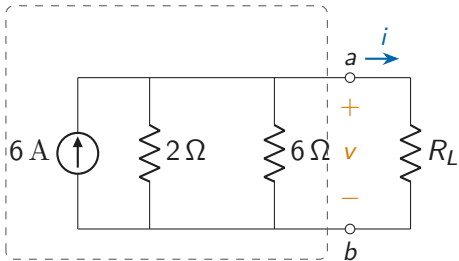
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

The first example, again



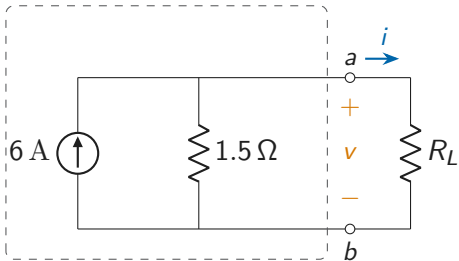
Now simplify only the part left of a–b; R_L stays.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $\quad(12 = 2 \times 6)$

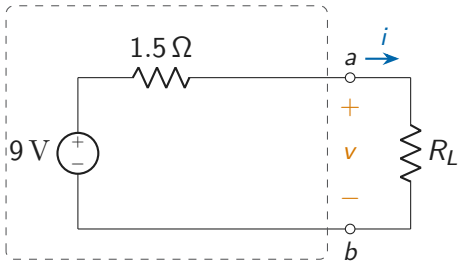
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$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

The first example, again

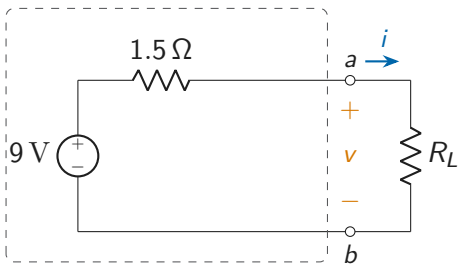


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The first example, again



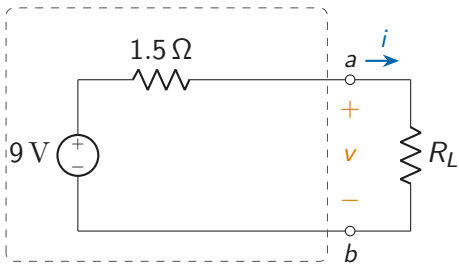
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Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \to 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \to 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

Solving LTI resistive circuits

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  1. node voltage analysis: the $n-1$ node voltages are the unknowns;
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  2. mesh current analysis: the $b-n+1$ mesh currents are the unknowns.
- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; if each circuit has a unique solution, nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.